OSP Cable - Optical Characteristics

This specification covers fibre characteristics for Single-mode Fibre ITU-T Rec. G.652.D and G.657.A1 and Multimode fibre OM1, OM3 and OM4 in 250 µm LT designs manufactured at the AFL Australia facility.

SINGLE-MODE FIBRE				
FIBRE SPECIFICATION (FIBRE CODE DESIGNATOR)		G.652.D (1D)	G.652.D (PREMIUM) (1E)	G.657.A1 (1F)
STRUCTURAL				
Typical Mode Field Diameter	@1310 nm	9.2 ± 0.4 µm	9.2 ± 0.4 µm	9.2 ± 0.4 µm
	@1550 nm	10.4 ± 0.5 µm	10.4 ± 0.5 µm	10.4 ± 0.5 µm
Cladding Diameter		125 ± 0.7 µm	125 ± 0.7 µm	125 ± 0.7 µm
Max Mode Field Concentricity Error		0.5 µm	0.5 µm	0.5 µm
Coating-Cladding Concentricity Error		≤ 12 µm	≤ 12 µm	≤ 12 µm
Cladding Non-circularity		≤ 0.7%	≤ 0.7%	≤ 0.7%
Fibre Coating Diameter (Coloured)		250 ± 10 µm	250 ± 10 µm	250 ± 10 µm
OPTICAL				
Max. Attenuation (Cabled)	@1310 nm	0.35 dB/km	0.33 dB/km	0.35 dB/km
	@1550 nm	0.21 dB/km	0.19 dB/km	0.21 dB/km
	@1625 nm	0.25 dB/km	0.25 dB/km	0.25 dB/km
Group Refractive Index	@1310 nm	1.4675	1.4675	1.4675
	@1550 nm	1.4681	1.4681	1.4681
	@1625 nm	1.4685	1.4685	1.4685
100 Turns, 50 mm Diameter	@1310 nm	≤ 0.01 dB	≤ 0.01 dB	≤ 0.01 dB
	@1550 nm	≤ 0.01 dB	≤ 0.01 dB	≤ 0.01 dB
	@1625 nm	≤ 0.01 dB	≤ 0.01 dB	≤ 0.01 dB
10 Turns, 30 mm Diameter	@1550 nm	-	-	≤ 0.05 dB
	@1625 nm	-	-	≤ 0.3 dB
1 Turn, 20 mm Diameter	@1550 nm	-	-	≤ 0.5 dB
	@1625 nm	-	-	≤ 1.5 dB
Max Cut-off Wavelength (λ_{cc})		1260 nm	1260 nm	1260 nm
Zero Disp^{n} Wavelength		1302 - 1324 nm	1302 - 1324 nm	1302 - 1324 nm
Slope at ZDW		$0.073 \leq S_0 \leq 0.092$ (ps/nm²$\cdot$km)	$0.073 \leq S_0 \leq 0.092$ (ps/nm²$\cdot$km)	$0.073 \leq S_0 \leq 0.092$ (ps/nm²$\cdot$km)
Typ Ch. Disp Coefficient	@1285 – 1330 nm	≤ 3.5 ps/(nm.km)	≤ 3.5 ps/(nm.km)	≤ 3.5 ps/(nm.km)
	@1550 nm (D_{1550})	$13.3 \leq D_{1550} \leq 18$ ps/(nm.km)	$13.3 \leq D_{1550} \leq 18$ ps/(nm.km)	$13.3 \leq D_{1550} \leq 18$ ps/(nm.km)
	@1625 nm (D_{1625})	$17.2 \leq D_{1625} \leq 22$ ps/(nm.km)	$17.2 \leq D_{1625} \leq 22$ ps/(nm.km)	$17.2 \leq D_{1625} \leq 22$ ps/(nm.km)
PMD Ind.Fibre/LDV		≤0.1/≤0.04 (ps/√km)	≤0.1/≤0.04 (ps/√km)	≤0.1/≤0.04 (ps/√km)
MECHANICAL				
Proof Test		≥ 1 %	≥ 1 %	≥ 1 %
Strip Force (F)		1.3 N ≤ F ≤ 8.9 N	1.3 N ≤ F ≤ 8.9 N	1.3 N ≤ F ≤ 8.9 N

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MULTIMODE FIBRE				
FIBRE SPECIFICATION (FIBRE CODE DESIGNATOR)		OM1 - 62.5 µm (62)	OM3 - 50 µm (53)	OM4 - 50 µm (55)
STRUCTURAL				
Typical Core Diameter		62.5 ± 2.5 µm	50.0 ± 2.5 µm	50.0 ± 2.5 µm
Max Core-Clad Conc Error		1 µm	1 µm	1 µm
Cladding Diameter		125 ± 1 µm	125 ± 1 µm	125 ± 1 µm
Fibre Coating Diameter (Natural)		242 ± 5 µm	242 ± 5 µm	242 ± 5 µm
MECHANICAL				
Proof Test		1 %	1 %	1 %
Strip Force (Avg)		1 - 3 N	1 - 3 N	1 - 3 N
OPTICAL				
Max. Attenuation (Cabled)	@850 nm	3.0 dB/km	3.0 dB/km	3.0 dB/km
	@1300 nm	1.0 dB/km	1.0 dB/km	1.0 dB/km
Min. Overfilled Bandwidth	@850 nm	200 MHz.km	1500 MHz.km (300 MHz.km 10GigEthernet)	3500 MHz.km (550 MHz.km 10GigEthernet)
	@1300 nm	500 MHz.km	500 MHz.km	500 MHz.km
Min. Effective Modal Bandwidth (EMB)		N/A	2000 MHz.km	4700 MHz.km
Numerical Aperture		0.275 ± 0.015	0.200 ± 0.015	0.200 ± 0.015
Group Refractive Index	@850 nm	1.496	1.482	1.482
	@1300 nm	1.491	1.477	1.477

FIBRE IDENTIFICATION	
Fibre Colour Sequence (1st 12f)	Blue, Orange, Green, Brown, Grey, White, Red, Black, Yellow, Violet, Pink, Aqua
Fibre Colour Sequence (2nd 12f)	Blue*, Orange*, Green*, Brown*, Grey*, White*, Red*, Natural*, Yellow*, Violet*, Pink*, Aqua* * With band mark

This datasheet is provided for guidance only and is subject to change without notice. Please ensure you have the most up to date information where parameters in this document are considered critical for your particular application.

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